Contents

Plan — from here to a fabricated board **1**
The decision that gates everything 1
Step 1 — Optional: re-polish on real copper 2
Step 2 — Fabrication outputs 2
Step 3 — Order the board 2
Later 2
Do not redo 3

Plan — from here to a fabricated board

Current state: `STATE.md`. Superseded plans: `docs/archive/`.

The design work is **done**, in the sense that the design space has been mapped and the remaining gap is understood rather than mysterious. What is left is a decision, then fabrication outputs.

The decision that gates everything

The specification cannot be met by this topology. Bandwidth and return loss are in direct opposition on every geometry parameter (see `STATE.md`, "the central finding"), and either can be met alone:

option	bandwidth	return loss	note
good match (current design)	316.6 MHz	**−10.76 dB**	2.2× the target band
good bandwidth	**145.6 MHz**	−1.75 dB	essentially unmatched
spec	143.5 MHz	−16 dB	not reachable together here

Decision taken 2026-09-03: the good-match end. The current design is that choice, optimised and confirmed.

Three ways forward, in the order I would consider them:

1. **Accept and fabricate.** Write the deviation down explicitly — a 316.6 MHz, −10.76 dB filter at 2.90 GHz — and order the board. Reasonable if the real requirement is "a working S-band cavity filter" rather than the literal numbers.
2. **Change the topology.** N=2 dual-mode has one internal coupling and one external coupling; that is not enough freedom to place two poles *and* match them. A cascade of two single-mode cavities has an extra degree

of freedom, at roughly 1.5× the board length. The parked cascade N=3 data is in `golden/cascade_N3/`.

3. **Relax the specification.** If 5 % FBW is a nice-to-have and return loss is the real requirement, the current design already satisfies the requirement that matters.

This is a user decision, not a technical one. Nothing further should be optimised until it is made.

Step 1 — Optional: re-polish on real copper

```
siw design --topology dualmode --N 2 \
    --seed a=57.158,w_iris=15.304,x_iris=28.685,y_wall=9.849,w_ext=16.352 \
    --span 2 --patience 14 --evals 12 --w-bw 0 --real-metal --device CPU --np 4
```

Probably not worth it. Real copper shifts the design by 0.32 dB of insertion loss and 0.21 dB of return loss without changing its shape, which is close to the coarse-mesh noise floor; twelve evaluations cost about 6.5 hours. The alternative is to accept the PEC-optimised geometry with the copper penalty applied — the numbers are already measured and in the golden entry.

Do **not** pass `--enig`: it implies ENIG + soldermask + silkscreen, all of which are unresolvable at this mesh density and destroy the response.

Step 2 — Fabrication outputs

- CST `.bas` export (`cst_export.py`) and a visual check in CST.
- Gerbers, with the **soldermask opened over the feed** (judgement call, see `STATE.md` — not a simulated result).
- JLCPCB DRC re-check against `config.py`'s recorded capabilities.
- ENIG finish, vias open/unfilled (tenting is not measurable and is electrically irrelevant at this via size).

Step 3 — Order the board

Only once the decision above is written down. Do not ship a board that silently misses spec.

Later

- **Paper** — "AI-Driven Design for SIW Filters". The strongest material is already collected, and most of it is negative or methodological:

  - a *structural* obstacle to black-box optimisation — an objective term that is discontinuous in the external coupling, plus the discovery that the objective's own bandwidth weight steered every search away from the best design;
  - **fifteen silent modelling bugs**, none of which raised an error;
  - the calibrate-and-invert failure with its mechanism (a dual-mode cavity's two resonators are two modes of one cavity, so no proxy can isolate one);
  - the bandwidth/match opposition as a measured property of the topology;
  - the limits of thin-layer modelling at practical mesh densities.
- **Mesh convergence study** for `fc` — never done; would put an error bar on every frequency in this project.

Do not redo

- **The fabrication stack study.** Soldermask, silkscreen and ENIG are unresolvable at this mesh density; the measured "penalties" scale with layer *thinness*, which is the signature of sliver elements, not physics. Re-running them produces the same meaningless numbers.
- **Volumetric lossy-conductor vias.** Same problem, and the hollow-barrel geometry is gated on them, which is why tented-vs-open is unanswerable.
- **Calibrate-and-invert on dual-mode.** The Qe proxy measures a different structure (3.01 GHz vs the design's 2.89 GHz); proxy loaded-Q 8.29 against the assembled filter's 8.66 proves it. Two coupled resonators cannot share the loaded Q of one.
- **Compact N=4 dual-mode cascade** — five mechanisms plus a joint search all hit the same ceiling. Raw data in `docs/geometry_exploration/`.
- **Lossy-conductor boundary conditions on internal surfaces** — Palace's `Conductivity` BC is exterior-only. It works correctly on the exterior ground planes, which is why the real-copper result is trustworthy.
- **A second launch definition.** There is one, in `topologies/launch.py`, plus an explicitly-named pad-only alternative. Forking it caused three separate bugs.